

Sakshi Bhatia

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PROFILE SUMMARY

I build manipulation systems that work on objects not seen before. My research couples foundation models with diffusion policies to achieve zero-shot generalization. I work across the full stack; building simulators, policy learning, and real robots.

EDUCATION

Northeastern University, Boston, MA <i>MS Robotics (Computer Vision, Reinforcement Learning, Geometrical Deep Learning)</i>	Sept 2024 – May 2026 GPA: 3.88/4.0
National Institute of Technology, Jalandhar, India <i>Bachelor of Technology (Soft Computing, Mechatronics)</i>	Aug 2018 – May 2022 GPA: 8.09/10

RESEARCH EXPERIENCE

Research Assistant , Dr. Robert Platt, NEU - Reproduced Diffusion Policy on RoboMimic and RoboCasa, built a GELLO from scratch for data collection on UR5 arm. - Designed a vision encoder for Diffusion Policy improving OOD generalization by 12.4% on Robomimic Lift with unseen objects; improves PushT success by 11%. Investigated various architectures and foundational models for policy learning. - Built RoboMimic2D, an open-source Gymnasium-based 2D manipulation benchmark with physics-based grasping, a gripper state machine, and zarr-backed replay buffer for imitation learning data collection.	Feb 2025 – Present
Research Intern , Dr. Anupam Yadav, NIT Jalandhar - Developed a Python package for 'Artificial Electric Field', a novel optimization algorithm, published as a pip-installable. - Converges to global minima/maxima for multi-dimensional, non-linear problems, while visualizing convergence plots in 2D.	July 2021 – Oct 2021

PUBLICATIONS

Enhancing Diffusion Policy Generalization through Structured Visual Representations (First Author, In Prep) Proposed a novel vision encoder trained under BC loss that improves OOD generalization by 12.4% across 12 unseen YCB objects. Integrates frozen SAM3 at inference for zero-shot segmentation without fine-tuning compute.	
Fuzzy-based techniques for clustering in wireless sensor networks , Wiley A review study published in International Journal of Communication Systems, https://doi.org/10.1002/dac.5583	July 2023

WORK EXPERIENCE

Teaching Assistant , EECE department, Northeastern University - Facilitated instruction for EECE5550: Mobile Robotics with Prof. Derya; graded assignments, conducted quizzes, and held office hours covering perception, localization, and control algorithms. - Assembled and deployed TurtleBot3 Burger robots, configuring, integrating, and testing hardware and software components.	Sep 2025 – Dec 2025
AI Product Analyst , Sprinklr, India Deployed a regex solution to automate harmful content removal on 200+ social channels, reducing manual work by 93%.	July 2022 – Jan 2023

PROJECTS

No reward long-horizon planning using latent world models - Implemented the Intrinsic Curiosity Module (ICM) with Deep Q-Network (DQN) for sparse environments. - Achieved an increase in success rate by 75% over vanilla DQN on sparse-reward long-horizon tasks.	Nov 2025
Augmented Reality C++ Pipeline - Developed an Augmented Reality application in C++ with real-time 3D spatial overlays and dynamic camera calibration. - Architected markerless target tracking through ORB matching and RANSAC homography, enhancing AR stability. - Integrated an OpenGL/GLFW rendering pipeline to overlay shaded 3D geometry onto live video streams.	March 2026

ACHIEVEMENTS

Best Android XR App with Gemini Live API Runner-up , by Google DeepMind – At MIT Reality Hack 2026
GATE Data Science and Artificial Intelligence : All India Rank 1243 / 39210 (96.83 %ile), by IISc Bengaluru 2024
Cleared Post Baccalaureate Fellowship at Robert Bosch's Centre for Data Science and AI- by IIT Madras 2023

SKILLS

Languages: Python, C++, SQL, MATLAB, Bash **Tools**: Git, Linux, NumPy, zarr
Robot Learning: Diffusion Policy, imitation learning, behavior cloning, reinforcement learning, vision-language-action (VLA) models, sim-to-real transfer, teleoperation **ML Frameworks**: PyTorch, TensorFlow, Diffusers, Transformers, Weights & Biases, CUDA, Docker **Vision**: OpenCV, SAM / SAM3, DINO, Grounded SAM, DepthPro, VLMs, diffusion models **Simulation & Robotics**: MuJoCo, RoboSuite, Robomimic, RoboCasa, Isaac Sim, Gymnasium, CARLA, Pymunk, Pygame, ROS / ROS 2